

Comprehensive Directives for the Artificial Intelligence Agent: Architecting a High-Fidelity, Rust-Based Cylindrical Magnetron Simulation

Introduction to the Computational Framework

The following directives constitute an exhaustive, expert-level specification for an Artificial Intelligence agent tasked with developing a high-fidelity, high-performance computational simulation of an electron trapped within the electromagnetic fields of a cylindrical magnetron. The foundational basis for this simulation is derived from a physical experimental study focusing on the determination of the initial velocity of electrons and the specific charge of the electron using the magnetron method.¹ However, the AI agent is instructed to transcend the simplified, non-relativistic, and space-charge-free approximations typical of secondary educational curricula. The resulting Rust-based simulation must serve as a rigorous digital twin of the physical apparatus, integrating advanced plasma physics phenomena, relativistic particle kinematics, and self-consistent electromagnetic field solvers.

Rust has been mandated as the language of implementation due to its unparalleled synthesis of memory safety, fearless concurrency, and zero-cost abstractions.² The simulation of non-neutral plasmas, particularly those exhibiting complex macroscopic fluid instabilities such as the diocotron effect, requires the tracking of millions of macro-particles alongside the continuous evaluation of multidimensional spatial grids. The AI agent must architect a codebase that leverages data-oriented design paradigms, utilizing the Rust ecosystem's premier numerical computing crates, to maximize cache locality and exploit single-instruction, multiple-data (SIMD) vectorization.³ By weaving the empirical baseline data with advanced kinetic models, the simulation will not only replicate the experimentally observed descending characteristics of the anode current but also unveil the underlying turbulent dynamics that classical analytical models obscure.

System Geometry and Physical Parameterization

The physical apparatus serving as the archetype for this simulation is the 2D2C (2D2S) direct-heated vacuum diode, situated within a homogeneous external solenoid.¹ The AI agent must establish a highly configurable, strictly typed configuration module within the Rust application that encodes these hardware specifications. The geometry of the interaction space is a coaxial cylindrical topology, defined by a central cathode wire and an encompassing solid cylindrical anode block.

The baseline parameters must be encoded as default values within the simulation environment, mapped to IEEE-754 64-bit floating-point types (f64) to ensure sufficient precision during the

calculation of steep potential gradients near the cathode surface.

Parameter Designation	Symbol	Nominal Value	Unit	Source
Anode Radius	R_a	5.1×10^{-3}	m	1
Cathode (Filament) Radius	r_c	6.25×10^{-5}	m	8
Solenoid Length	l	0.167	m	1
Solenoid Diameter	d	0.062	m	1
Solenoid Turn Count	N	2300	-	1
Anode Voltage Range	U_a	40 – (max 140)	V	1
Filament Heating Voltage	U_f	1.2 –	V	6
Maximum Anode Current	$I_{a,max}$	40	mA	6
Nominal Initial Velocity	v_0	$1.1 \times$	m/s	1
Electron Specific Charge	e/m	$1.9 \times$	C/kg	1

The AI agent must programmatically construct the simulation boundary conditions based on this table. The cathode of the 2D2S is a thorium-oxide coated tungsten filament designed for

direct heating.¹⁰ The agent must recognize that direct heating introduces a fundamental asymmetry into the idealized 2D cylindrical model. Because the heating voltage U_f (approximately 1.2 to 1.7 Volts) is applied directly across the ends of the cathode wire to induce thermionic emission, the electric potential along the surface of the cathode is not uniform.⁶ It varies linearly from 0 Volts at the grounded terminal to U_f at the polarized terminal.

For an experimental anode voltage U_a operating in the 40 Volt regime, a 1.5 Volt longitudinal differential along the z-axis of the cathode represents a substantial gradient.¹ This gradient establishes an axial electric field component E_z in addition to the primary radial field E_r . The presence of E_z dictates that the simulation cannot be reduced to a purely two-dimensional planar cross-section without sacrificing physical fidelity. The electrons will experience an axial drift force, necessitating a full three-dimensional coordinate tracking system. The Rust field solver must be instructed to compute the electrostatic potential $V(r, z, \theta)$ iteratively, applying Dirichlet boundary conditions $V(R_a, z, \theta) = U_a$ at the anode and $V(r_c, z, \theta) = z \cdot (U_f/l_c)$ along the length l_c of the cathode.

Baseline Analytical Framework and The Hull Cutoff

The core operational principle of the magnetron diode, as outlined in the baseline empirical study, revolves around the transition of electron trajectories from radial transit to azimuthal circulation under the influence of crossed electric and magnetic fields.¹ The AI agent must implement synthetic diagnostic routines that parallel the mathematical derivations used by the students and researchers to extract the specific charge of the electron (e/m) and the initial thermal velocity (v_0).

In an idealized vacuum state devoid of space charge, the electric field is purely radial, and the external magnetic field provided by the solenoid is purely axial. The magnetic field B generated by the solenoid at its center is approximated by the finite-solenoid equation incorporating the length l and diameter d :

$$B = \frac{\mu_0 N I_c}{\sqrt{l^2 + d^2}}$$

where I_c is the current passing through the solenoid windings.¹ The AI agent must implement a continuous spatial evaluation function for the magnetic field vector $\mathbf{B}(r, z)$ that accounts for the magnetic field fringing effects near the axial extremities of the solenoid, utilizing a

complete Biot-Savart integration over the coil geometry if highly accurate off-axis fields are required.

The primary metric of interest is the Hull cutoff condition.¹⁵ The Lorentz force governs the acceleration of the electron:

$$\frac{d\mathbf{p}}{dt} = -e(\mathbf{E} + \mathbf{v} \times \mathbf{B})$$

Due to the cylindrical symmetry, the system conserves the canonical angular momentum P_θ , as formalized by Busch's Theorem.¹⁵ For an electron emitted from the cathode surface at $r = r_c$ with an assumed initial angular velocity of zero, the canonical angular momentum is defined solely by the magnetic vector potential at the cathode.¹⁷ As the electron moves radially outward to a generic radius r , the conservation law dictates:

$$mr^2\dot{\theta} - \frac{1}{2}eBr^2 = -\frac{1}{2}eBr_c^2$$

Rearranging this expression isolates the angular velocity $\dot{\theta}$ of the electron at any point in the interaction space:

$$\dot{\theta} = \frac{eB}{2m} \left(1 - \frac{r_c^2}{r^2} \right)$$

The Hull cutoff represents the critical magnetic field threshold B_k at which the electron's trajectory becomes tangent to the anode surface.¹ At this precise condition, the radial velocity $\dot{r}$ drops to zero at $r = R_a$, meaning all kinetic energy is concentrated in the azimuthal direction. Applying the conservation of energy, where the work done by the electric field eU_a equals the final kinetic energy, yields¹⁶:

$$eU_a = \frac{1}{2}m(R_a\dot{\theta})^2$$

Substituting the angular velocity at $r = R_a$ into the energy equation provides the exact Hull cutoff magnetic field¹⁶:

$$B_k = \sqrt{\frac{8mU_a}{eR_a^2 \left(1 - \frac{r_c^2}{R_a^2} \right)^2}}$$

Given that the radius of the cathode r_c in the 2D2S diode (6.25×10^{-5} m) is approximately two orders of magnitude smaller than the anode radius R_a (5.1×10^{-3} m), the term $(r_c/R_a)^2$ is frequently negligible.¹ This allows for the simplified determination of the specific charge utilized in standard laboratory practices:

$$\frac{e}{m} = \frac{8U_a}{B_k^2 R_a^2}$$

However, the uploaded research explicitly corrects this idealized model by accounting for the initial thermal velocity v_0 of the electrons resulting from thermionic emission.¹ The AI agent must design the simulation to initialize macro-particles with a Maxwell-Boltzmann velocity distribution characteristic of the cathode's operating temperature. The modified energy conservation equation becomes¹:

$$\frac{mv^2}{2} = eU_a + \frac{mv_0^2}{2}$$

By combining this with the tangential velocity condition at the cutoff radius, the relationship between the square of the critical magnetic field and the applied anode voltage transforms into a linear equation of the form¹:

$$B_k^2 = \alpha U_a + \beta$$

where the slope $\alpha = \frac{8}{(e/m)R_a^2}$ enables the extraction of the specific charge, and the intercept $\beta = \frac{8v_0^2}{(e/m)^2 R_a^2}$ permits the isolation of the initial velocity.¹ The AI agent is instructed to build a synthetic diagnostic module within the Rust application that automatically conducts a parameter sweep over multiple anode voltages (U_a in V), observes the descent of the synthetic anode current, numerically computes the inflection points to identify I_{ck} and B_k , and performs a least-squares linear regression on the $B_k^2(U_a)$ data. The success criterion for the simulation's foundational physics engine is the autonomous recovery of the experimentally determined values: $e/m \approx 1.9 \times 10^{11}$ C/kg and $v_0 \approx 1.1 \times 10^6$ m/s.¹

Advanced Physics Subtleties I: Space-Charge Limited Flow and the Langmuir-Blodgett Law

The analytical framework described above relies on the assumption that electrons traverse the vacuum gap as isolated tracer particles, uninfluenced by the electric fields generated by their

peers. The AI agent must recognize this as a severe physical oversimplification. In operational magnetrons, thermionic emission produces a dense electron cloud near the cathode. This negative space charge acts to screen the applied positive potential from the anode, depressing the local electric field.²¹

When the temperature of the cathode is sufficiently high, the abundance of emitted electrons

drives the electric field precisely at the cathode surface to zero ($E(r_c) = 0$). This state is known as Space-Charge Limited (SCL) flow, and it represents the fundamental boundary condition for the particle injection module in the Rust simulation.²¹ The AI agent must implement an injection algorithm that dynamically introduces macro-particles into the simulation domain only until the local electric field at the boundary vanishes, thereby self-regulating the emission current.²⁵

For a planar geometry, SCL flow is governed by the Child-Langmuir law. However, the coaxial cylindrical geometry of the 2D2S magnetron imposes a divergent electric field profile,

necessitating the use of the Langmuir-Blodgett law.²⁶ The space-charge limited current I_{SCL} per unit length l in a cylindrical diode is given by ²⁸:

$$I_{SCL} = \frac{8\pi\epsilon_0}{9} \sqrt{\frac{2e}{m}} \frac{l U_a^{3/2}}{r \beta^2}$$

The parameter β is a highly complex, nonlinear geometric correction factor derived from the series expansion of the Poisson equation in cylindrical coordinates.²⁷ The AI agent must encode

the determination of β to properly benchmark the continuous fluid limits of the PIC simulation.

The series expansion, centered on $\gamma = \ln(r/r_c)$, is formulated as ²⁹:

$$\beta = \gamma - \frac{2}{5}\gamma^2 + \frac{11}{120}\gamma^3 - \frac{47}{3300}\gamma^4 + \dots$$

To capture these effects dynamically, the Rust architecture must couple the kinematic particle tracker with a self-consistent Poisson solver.²² At each computational time step, the position of every macro-particle is mapped onto a discrete spatial grid using a charge-conserving weighting algorithm, such as Cloud-In-Cell (CIC) or area-weighting.³⁰ This generates a

volumetric charge density field $\rho(r, z, \theta)$. The agent must then instruct the software to solve the Poisson equation:

$$\nabla^2 V = -\frac{\rho}{\epsilon_0}$$

The resulting electrostatic potential grid is differentiated to update the spatial electric field array, which is subsequently interpolated back to the exact continuous coordinates of each

macro-particle to compute the Lorentz force for the next kinetic update.³⁰ The necessity of resolving the Debye length (to prevent artificial numerical grid heating) and the plasma frequency (to ensure temporal stability) places strict constraints on the spatial and temporal resolution parameters the AI agent must select for the simulation mesh.³¹

Advanced Physics Subtleties II: Shear Flow and the Diocotron Instability

The coupling of the individual particle dynamics with the macroscopic space-charge field leads directly to the manifestation of turbulent plasma phenomena, the most critical of which for this simulation is the diocotron instability.³² This instability is the physical mechanism responsible for

the gradual, smooth descent of the anode current I_a observed in the empirical data¹, which sharply contradicts the instantaneous drop to zero predicted by the idealized Hull cutoff equation.³⁵

When the applied magnetic field exceeds the critical threshold ($B > B_k$), the electrons are theoretically confined to a rotating, cylindrical cloud surrounding the cathode, known as a Brillouin flow.²⁴ Within this cloud, the crossed radial electric field and axial magnetic field induce

an azimuthal $\mathbf{E} \times \mathbf{B}$ drift velocity.³² However, because the electric field intensity and the space-charge density vary significantly with the radius r , the azimuthal drift velocity is not constant; it exhibits intense radial shear ($d\omega/dr \neq 0$).³³

The AI agent must be programmed to understand that this differential rotation renders the electron plasma analogous to an inviscid fluid experiencing shear, making it highly susceptible to Kelvin-Helmholtz-like fluid dynamic instabilities.³³ Minor random fluctuations in the local charge density—arising from thermal noise or discrete emission events—grow exponentially over time. This growth breaks the azimuthal symmetry of the electron cloud, leading to the formation of distinct, macroscopic electron vortices or "spokes" that rotate around the cathode.³¹

The characteristic frequency of this wave motion scales with the $\mathbf{E} \times \mathbf{B}$ drift and the azimuthal mode number m .³² The presence of these diocotron modes generates localized, transient azimuthal electric fields ($E_\theta \neq 0$). When an electron is subjected to an azimuthal electric field in the presence of an axial magnetic field, it experiences a radial $\mathbf{E}_\theta \times \mathbf{B}_z$ drift force.³¹ This causes anomalous cross-field transport, actively pumping electrons radially outward toward the anode even when the average magnetic field heavily exceeds the Hull cutoff.³⁵

Therefore, the AI agent must construct the simulation space with full 3D Cartesian or 3D

cylindrical (r, θ, z) degrees of freedom. Enforcing a 1D radial or 2D axisymmetric constraint would artificially suppress the azimuthal symmetry breaking, effectively preventing the diocotron instability from developing, and resulting in an inaccurate, infinitely sharp cutoff curve that fails to match the uploaded Ukrainian research data.¹

Algorithmic Framework: The Relativistic Boris Integrator

The temporal integration of the electron trajectories demands an algorithm capable of preserving the phase-space geometry over thousands of cyclotron periods. The AI agent must critically evaluate the Rust crate ecosystem and avoid relying on standard explicit Runge-Kutta (RK4) solvers for the particle kinematics, despite their availability in highly optimized crates like `diffsol` and `differential-equations`.² Standard Runge-Kutta methods are not symplectic; they introduce artificial energy dissipation or unbounded numerical heating when tracking cyclic orbits in strong magnetic fields, eventually causing the simulated electrons to spiral inwards or outwards unphysically.³⁰

Instead, the AI agent must explicitly code a custom volume-preserving integrator: the Relativistic Boris Algorithm (or Boris Pusher).⁴⁴ The Boris method is a second-order, time-centered leapfrog scheme that perfectly preserves the phase space volume and the

crucial $\mathbf{E} \times \mathbf{B}$ drift velocity, rendering it the de facto standard in plasma physics for its exceptional long-term stability.⁴¹

Given the potential for anode voltages scaling up to 140 Volts⁶ and the immense acceleration forces within the microscopic vacuum gap, the electrons can achieve velocities where relativistic mass corrections become noticeable, altering the cyclotron frequency and the Hull cutoff boundary.²⁴ The relativistic equations of motion are defined in terms of the proper

momentum $\mathbf{u} = \gamma \mathbf{v}$, where $\gamma = \sqrt{1 + (\mathbf{u}/c)^2}$ is the Lorentz factor.⁴⁶

The AI agent must construct the Rust integration loop to advance the momentum from step $n - 1/2$ to $n + 1/2$ over a discrete time step Δt through a meticulously split four-stage process⁴³:

1. First Half-Electric Acceleration:

The particle is accelerated for half a time step using solely the local electric field $\mathbf{E}^n$, yielding an intermediate momentum $\mathbf{u}^-$.

$$\mathbf{u}^- = \mathbf{u}^{n-1/2} + \frac{q\Delta t}{2m_e} \mathbf{E}^n$$

2. Relativistic Factor Computation: Because magnetic fields perform no work, the kinetic

energy—and therefore the Lorentz factor γ —remains perfectly constant during the

magnetic rotation phase.⁵² The agent must compute this conserved γ^n using the intermediate momentum:

$$\gamma^n = \sqrt{1 + \frac{\mathbf{u}^- \cdot \mathbf{u}^-}{c^2}}$$

3. **Magnetic Rotation via Boris Transformation:** The momentum vector must be rotated around the magnetic field axis $\mathbf{B}^n$. Solving the implicit rotation directly requires computationally expensive 3x3 matrix inversions.⁴⁶ The genius of the Boris algorithm is the use of intermediate vectors to perform a pure geometric rotation via cross products.⁵³ The agent must define a rotation vector $\mathbf{t}$ and a scaling vector $\mathbf{s}$:

$$\mathbf{t} = \frac{q\Delta t}{2m_e\gamma^n} \mathbf{B}^n$$

$$\mathbf{s} = \frac{2\mathbf{t}}{1 + |\mathbf{t}|^2}$$

The rotation is then executed seamlessly:

$$\mathbf{u}' = \mathbf{u}^- + (\mathbf{u}^- \times \mathbf{t})$$

$$\mathbf{u}^+ = \mathbf{u}' + (\mathbf{u}' \times \mathbf{s})$$

4. **Second Half-Electric Acceleration:**

The remaining half of the electric impulse is added to finalize the momentum update:

$$\mathbf{u}^{n+1/2} = \mathbf{u}^+ + \frac{q\Delta t}{2m_e} \mathbf{E}^n$$

Finally, the positional coordinates are updated using the time-centered, relativistically corrected velocity:

$$\mathbf{x}^{n+1} = \mathbf{x}^n + \frac{\mathbf{u}^{n+1/2}}{\gamma^{n+1/2}} \Delta t$$

The AI agent is directed to abstract this algorithm into a reusable Rust trait, for example, trait ParticlePusher, ensuring it is decoupled from the specific electromagnetic field representations to allow for modular unit testing against analytical single-particle orbits.⁵⁴

Rust Implementation Directives: Memory Architecture and Crate Ecosystem

The computational bottleneck in an N -body Particle-In-Cell simulation is rarely the arithmetic logic unit (ALU); it is almost universally memory bandwidth and cache latency.⁵⁵ The AI agent must strictly avoid Object-Oriented paradigms. Allocating individual particles as structs (e.g., `struct Electron { x: f64, y: f64, ... }`) stored within a standard `Vec<Electron>` results in an Array of

Structures (AoS) memory layout. When the CPU attempts to update the $\mathbf{x}$ -coordinates of a million electrons simultaneously, the AoS layout forces the cache to load unnecessary data (like velocities and mass identifiers), causing devastating cache thrashing and preventing SIMD auto-vectorization.

The architecture must be constructed using Data-Oriented Design (DoD) principles, implementing a Structure of Arrays (SoA) layout.³ To achieve this, the agent is mandated to utilize the `ndarray` crate.³ The phase space of the entire electron swarm must be represented by dense, contiguous multidimensional arrays.⁴

For example, the kinematic state should be encapsulated as follows:

```
Rust
use ndarray::Array2;

pub struct PhaseSpace
/// Shape (3, N) containing x, y, z coordinates
pub positions: Array2<f64>
/// Shape (3, N) containing relativistic momentum ux, uy, uz
pub momenta: Array2<f64>
}
```

By organizing the data such that all $\mathbf{x}$ -coordinates occupy a single contiguous block of RAM, the CPU prefetcher can stream the data flawlessly into the L1 cache. The Boris integration steps detailed in the previous section must be implemented using `ndarray::Zip` operations.⁴

Furthermore, the agent must leverage the `rayon` crate to convert these operations into parallel iterators, instantly distributing the workload across all available CPU cores with minimal boilerplate code.

Fluid and Circuit Modeling Integration

While the Boris algorithm manages the discrete kinetic particle tracking, the simulation must also account for macroscopic boundaries, such as the temporal evolution of the external solenoid current I_c or the RLC circuit envelope of the power supply. These macroscopic systems are best modeled as continuous ordinary differential equations (ODEs). For these systems, the AI agent is instructed to deploy the `diffsol`² or `differential-equations`⁴⁰ crates. `diffsol` is highly recommended for solving stiff systems using Backward Difference Formulae (BDF) or Singly Diagonally Implicit Runge-Kutta (SDIRK) methods.³⁹ The architecture must segregate the fast-timescale kinetic tracking (femtosecond steps required to resolve the plasma frequency) from the slow-timescale circuit evolution (microsecond steps), coupling them only at defined synchronization barriers using an overarching supervisor loop.

Subsystem	Mathematical Domain	Rust Integration Tool/Algorithm	Justification
Particle Kinematics	Discrete, Non-linear	Custom Relativistic Boris Pusher	Preserves phase-space volume and $\mathbf{E} \times$ drift. ⁴¹
Memory Layout	Structure of Arrays (SoA)	<code>ndarray</code> , <code>rayon</code> crates	Maximizes SIMD vectorization and L1 cache hits. ³
External Circuits	Continuous, Stiff ODEs	<code>diffsol</code> or <code>differential-equations</code>	Handles stiff transients during voltage ramp-up. ³⁹
Field Solving	Poisson Equation	Custom Finite Difference/Multigrid	Required for self-consistent space charge limits. ²⁵

Floating-Point Integrity and Normalization

A critical software engineering directive for the AI agent concerns floating-point precision loss. The simulation deals with an extreme disparity in physical scales: macroscopic dimensions are on the order of millimeters (10^{-3}), while the electron mass is incredibly small (9.109×10^{-31} kg) and the fundamental charge is similarly miniscule (1.602×10^{-19} C). Directly inserting these SI units into 64-bit floating-point registers during the evaluation of the Lorentz force will trigger catastrophic arithmetic underflow or precision truncation.

The agent must architect the simulation engine entirely in dimensionless units. The fundamental equations must be mathematically normalized before a single line of Rust is written.

- Spatial dimensions must be normalized by the anode radius: $\tilde{x} = x/R_a$.
- Time must be normalized by the inverse of the characteristic electron cyclotron frequency $\omega_c = eB/m_e$ or the plasma frequency ω_p : $\tilde{t} = t \cdot \omega_c$.
- Velocities must be normalized by the speed of light c : $\tilde{v} = v/c$.

By mapping all operational parameters to $\mathcal{O}(1)$ dimensionless variables, the IEEE-754 precision is completely preserved throughout the millions of iterative integration steps, and the convergence rate of the iterative Poisson solvers is dramatically accelerated.

Synthetic Diagnostics and Automated Validation Pipeline

The ultimate metric of success for this computational framework is its ability to serve as a digital twin capable of autonomously replicating the baseline empirical results found in the Ukrainian research document.¹ To achieve this, the AI agent must construct a sophisticated synthetic diagnostic module that operates transparently alongside the physics engine.

1. **The Virtual Microammeter:** The simulation must establish a boundary-flux counting algorithm at the anode radius $r = R_a$. Every macro-particle that crosses this spatial threshold within a given time step Δt must be logged, absorbed, and removed from the active ndarray pool. The flux is then multiplied by the macro-particle weight (the number of physical electrons represented by one simulated entity) and the elementary charge to compute a time-averaged synthetic anode current I_a .
2. **Descending Characteristic Extraction:** The supervisor loop must be programmed to execute an automated parametric sweep mirroring the physical experiment.¹ The anode voltage U_a is clamped to a fixed value (e.g., 40, 50, 60, 70, 80 V). The solenoid current I_c is incrementally stepped upwards. At each step, the simulation is allowed to reach a quasi-steady state (after the initial transient SCL expansion), and the synthetic I_a is recorded. The output dataset $I_a(I_c)$ will exhibit a stable plateau followed by a distinct descending curve smoothed by the diocotron instabilities.¹
3. **Autonomous Inflection Analytics:** To mathematically isolate the critical Hull cutoff magnetic field B_k , the software must process the descending curve $I_a(I_c)$ to find the point of maximum slope. The agent is directed to implement a numerical differentiation routine to calculate the second derivative of the array and isolate the root, thereby

extracting the critical current I_{ck} automatically, removing all human subjectivity from the measurement.¹

4. **Linear Regression for Subatomic Parameters:** Following the sweeps across the full voltage range, the diagnostic module must construct the B_k^2 versus U_a array. The agent will utilize standard Rust statistical libraries to perform a least-squares linear regression.¹ By analyzing the resultant slope α and the y-intercept β , the simulation will output its calculated internal specific charge e/m and the initial thermal velocity v_0 . The validation of the entire Rust codebase, the Boris integrator, the Poisson solver, and the SCL injection logic is strictly conditional upon this final automated output. The synthetic parameters derived by the AI agent's code must match the input physics—specifically, yielding an extracted initial velocity of $v_0 \approx 1.1 \times 10^6$ m/s and a specific charge of $e/m \approx 1.9 \times 10^{11}$ C/kg with a relative numerical error margin of less than 1% against the baseline experimental data.¹ Through this rigorous synthesis of data-oriented systems programming and advanced plasma physics, the AI agent will deliver a definitive computational laboratory for cylindrical magnetron dynamics.

Cytowane prace

1. Дослідницька робота.docx
2. martinjrobins/diffsol: ODE solver library in Rust - GitHub, otwierano: czerwca 26, 2026, <https://github.com/martinjrobins/diffsol>
3. ndarray - Rust - Docs.rs, otwierano: czerwca 26, 2026, <https://docs.rs/ndarray/latest/ndarray/>
4. ndarray - crates.io: Rust Package Registry, otwierano: czerwca 26, 2026, <https://crates.io/crates/ndarray>
5. Chapter 1 | Computational Physics via Rust - RantAI, otwierano: czerwca 26, 2026, <https://cpvr.rantai.dev/docs/part-i/chapter-1/>
6. 2Д2С - Lampes & Tubes, otwierano: czerwca 26, 2026, <https://lampes-et-tubes.info/mwot/2D2S.pdf>
7. Радиолампа 2Д2С купить по низкой цене в Москве - РадиоТехИндустрия, otwierano: czerwca 26, 2026, https://www.r-t-i.ru/product_info/11746/
8. КУРС ФИЗИКИ, otwierano: czerwca 26, 2026, [https://publ.lib.ru/ARCHIVES/B/"Berkleevskiy_kurs_fiziki"_\(seriya\)/%C1%E5%F0%E A%EB%E5%E5%E2%F1%EA%E8%E9%20%EA%F3%F0%F1%20%F4%E8%E7%E8% EA%E8.%20%D2%EE%EC%2006.%20_%CF%EE%F0%F2%E8%F1%20%CO._%20 %D4%E8%E7%E8%F7%E5%F1%EA%E0%FF%20%EB%E0%E1%EE%F0%E0%F2%E E%F0%E8%FF.\(1972\).pdf](https://publ.lib.ru/ARCHIVES/B/)
9. ШУМОВОЙ ДИОД ТИПА 2Д2С - retrostore.eu, otwierano: czerwca 26, 2026, https://retrostore.eu/files/eshop/download/2d2s_data.pdf
10. SVETLANA 2D2S Noise Diode - Lampes & Tubes, otwierano: czerwca 26, 2026,

- <https://lampes-et-tubes.info/mwot/mw204.php?l=d>
11. 2D2S MEASURING NOISE DIODE VINTAGE TUBE (for noise generator) 2pcs | eBay, otwierano: czerwca 26, 2026, <https://www.ebay.com/itm/325443563802>
 12. Magnetron - Radartutorial.eu, otwierano: czerwca 26, 2026, <https://www.radartutorial.eu/08.transmitters/Magnetron.en.html>
 13. Filament X-ray Tube Current Control Method Using Indirect Filament Temperature Estimation - MDPI, otwierano: czerwca 26, 2026, <https://www.mdpi.com/2076-3417/11/22/10588>
 14. What dictates the current drawn by a magnetron's anode? - Electronics Stack Exchange, otwierano: czerwca 26, 2026, <https://electronics.stackexchange.com/questions/654766/what-dictates-the-current-drawn-by-a-magnetrons-anode>
 15. The Hull Cutoff condition for magnetic insulation in crossed-field electron devices in the presence of a slow-wave structure | Journal of Applied Physics | AIP Publishing, otwierano: czerwca 26, 2026, <https://pubs.aip.org/aip/jap/article/135/9/093303/3268665/The-Hull-Cutoff-condition-for-magnetic-insulation>
 16. Hull Cutoff Voltage in Cylindrical Magnetron | PDF | Electron | Physics - Scribd, otwierano: czerwca 26, 2026, <https://www.scribd.com/doc/106013937/Hull-Cut-Off-Equation-Electron-Motion-in-Cylindrical-Magnetron>
 17. EXTENSION OF BUSCH'S THEOREM TO PARTICLE BEAMS - JACoW / Proceedings, otwierano: czerwca 26, 2026, <https://proceedings.jacow.org/linac2018/papers/tupo078.pdf>
 18. Electron Sources: Single Particle Dynamics, Space Charge Limited Emission - classe, otwierano: czerwca 26, 2026, <https://www.classe.cornell.edu/~ib38/uspas08/lec3.pdf>
 19. Chapter 12 - CROSSED-FIELD DEVICES - Western Electric, otwierano: czerwca 26, 2026, <https://western-electric.squarespace.com/s/Chapter-12-CROSS-FIELD-DEVICES.pdf>
 20. Magnetron: Hull's cut off voltage equation - YouTube, otwierano: czerwca 26, 2026, <https://www.youtube.com/watch?v=DKxcPN9yxvI>
 21. MULTIDIMENSIONAL SPACE CHARGE LIMITED CURRENT THEORIES FOR CYLINDRICAL, TIP-TO-TIP, AND THICK PLANAR ELECTRODES - Purdue University Graduate School research repository, otwierano: czerwca 26, 2026, <https://hammer.purdue.edu/ndownloader/files/54146636>
 22. A Multi-Dimensional Child-Langmuir Law for Any Diode Geometry - ScholarWorks, otwierano: czerwca 26, 2026, https://scholarworks.boisestate.edu/cgi/viewcontent.cgi?article=1502&context=electrical_facpubs
 23. 100 years of the physics of diodes - AIP Publishing, otwierano: czerwca 26, 2026, <https://pubs.aip.org/aip/apr/article/4/1/011304/123867/100-years-of-the-physics-of-diodes>
 24. Numerical simulation of a relativistic magnetron using a uid electron model -

- OSTI, otwierano: czerwca 26, 2026, <https://www.osti.gov/servlets/purl/1888553>
25. Novel Scaling Laws for the Langmuir-Blodgett Solutions in Cylindrical and Spherical Diodes - Plasma, Pulsed Power, and Microwave Laboratory, otwierano: czerwca 26, 2026, <https://plasmabay.engin.umich.edu/wp-content/uploads/sites/281/2023/10/Zhu-2013-Novel-Scaling-Laws-for-the-Langmuir-Blodgett-Solutions-in-Cylindrical-and-Spherical-Diodes.pdf>
26. Space-charge-limited current density for nonplanar diodes with monoenergetic emission using Lie-point symmetries - arXiv, otwierano: czerwca 26, 2026, <https://arxiv.org/pdf/2204.01507>
27. On relativistic space charge limited current in planar, cylindrical, and spherical diodes - MSU Engineers, otwierano: czerwca 26, 2026, <https://www.egr.msu.edu/~pz/rd.pdf>
28. 2e冊1/2 - AIP Publishing, otwierano: czerwca 26, 2026, https://pubs.aip.org/aip/pop/article-pdf/11/6/3278/19169285/3278_1_online.pdf
29. CALCULATING SPACE-CHARGE-LIMITED CURRENT DENSITY IN NONPLANAR AND MULTI-DIMENSIONAL DIODES - Purdue University Graduate School, otwierano: czerwca 26, 2026, <https://hammer.purdue.edu/ndownloader/files/45997692>
30. Chapter 32 | Computational Physics via Rust - RantAI, otwierano: czerwca 26, 2026, <https://cpvr.rantai.dev/docs/part-vii/chapter-32/>
31. The magnetron instability in a pulsar's cylindrical electrosphere. - arXiv, otwierano: czerwca 26, 2026, <https://arxiv.org/pdf/0711.2416>
32. The Diocotron Instability in the Trapped Electrons Experiment T-REX and its Relevance to Electron Clouds in Gyrotron Guns - arXiv, otwierano: czerwca 26, 2026, <https://arxiv.org/html/2606.02217v1>
33. Diocotron instability | Plasma-Universe.com, otwierano: czerwca 26, 2026, <https://www.plasma-universe.com/diocotron-instability/>
34. THE DIOCOTRON INSTABILITY IN A CYLINDRICAL GEOMETRY R. H. Levy RESEARCH REPORT 202 December 7964 AERONAUTICS AND SPACE ADMINSTR, otwierano: czerwca 26, 2026, <https://ntrs.nasa.gov/api/citations/19650025523/downloads/19650025523.pdf>
35. Cut-off Characteristics of a Cylindrical Magnetron , otwierano: czerwca 26, 2026, <https://www.tandfonline.com/doi/pdf/10.1080/00207216608937929>
36. Stability of Brillouin flow in the presence of slow-wave structure, otwierano: czerwca 26, 2026, <https://plasmabay.engin.umich.edu/wp-content/uploads/sites/281/2021/08/Simon-2016-Stability-of-Brillouin-flow-in-the-presence-of-slow-wave-structure.pdf>
37. Simulations of the instability of the m=1 self-shielding diocotron mode in finite-length nonneutral plasmas - BYU Physics and Astronomy, otwierano: czerwca 26, 2026, <https://physics.byu.edu/docs/publication/2430>
38. Theory of Cross Field Devices and a Comparative Study of Other Radiation Sources. - DTIC, otwierano: czerwca 26, 2026, <https://apps.dtic.mil/sti/tr/pdf/ADA183389.pdf>
39. diffsol - Rust - Docs.rs, otwierano: czerwca 26, 2026, <https://docs.rs/diffsol>

40. differential-equations - crates.io: Rust Package Registry, otwierano: czerwca 26, 2026, <https://crates.io/crates/differential-equations>
41. PPPL- 4872 - 4872 Why is Boris Algorithm So Good?, otwierano: czerwca 26, 2026, https://bp-pub.pppl.gov/pub_report/2013/PPPL-4872.pdf
42. An improved Boris algorithm for charge particle orbit in tokamak plasmas, otwierano: czerwca 26, 2026, <http://pst.hfcas.ac.cn/en/article/pdf/preview/10.1088/2058-6272/adfcb4.pdf>
43. A Comparative Analysis of Relativistic Particle Pushers vis-à-vis Computation Time & Accuracy - arXiv, otwierano: czerwca 26, 2026, <https://arxiv.org/html/2410.03352v1>
44. An Improved Boris Algorithm for Charge Particle Orbit in Tokamak Plasmas - arXiv, otwierano: czerwca 26, 2026, <https://arxiv.org/abs/2504.01314>
45. The Boris algorithm for numerically tracing non-relativistic charged particles in electromagnetic fields, written in C, matlab, and python - GitHub, otwierano: czerwca 26, 2026, <https://github.com/iwhoppock/boris-algorithm>
46. 1 An Analytic Boris Pusher for Plasma Simulation Viktor K. Decyk, Warren B. Mori, and Fei Li Department of Physics and Astronomy - arXiv, otwierano: czerwca 26, 2026, <https://arxiv.org/pdf/2204.13563>
47. Simulation of beams or plasmas crossing at relativistic velocity. Abstract - OSTI, otwierano: czerwca 26, 2026, <https://www.osti.gov/servlets/purl/926621>
48. picFoam: An OpenFOAM based electrostatic Particle-in-Cell solver | Request PDF, otwierano: czerwca 26, 2026, https://www.researchgate.net/publication/348823426_picFoam_An_OpenFOAM_based_electrostatic_Particle-in-Cell_solver
49. Particle Simulation and Optimization of a Relativistic Magnetron for HPM Applications - <https://uu.diva-portal.org>, otwierano: czerwca 26, 2026, <https://uu.diva-portal.org/smash/get/diva2:1678756/FULLTEXT01.pdf>
50. 9.6: Relativistic Particles in Electric and Magnetic Fields - Physics LibreTexts, otwierano: czerwca 26, 2026, [https://phys.libretexts.org/Bookshelves/Electricity_and_Magnetism/Essential_Graduate_Physics_-_Classical_Electrodynamics_\(Likharev\)/09%3A_Special_Relativity/9.06%3A_Relativistic_Particles_in_Electric_and_Magnetic_Fields](https://phys.libretexts.org/Bookshelves/Electricity_and_Magnetism/Essential_Graduate_Physics_-_Classical_Electrodynamics_(Likharev)/09%3A_Special_Relativity/9.06%3A_Relativistic_Particles_in_Electric_and_Magnetic_Fields)
51. Modelling and Optimisation of Relativistic Magnetron with Transparent Cathode - <https://uu.diva-portal.org>, otwierano: czerwca 26, 2026, <https://uu.diva-portal.org/smash/get/diva2:1755953/FULLTEXT01.pdf>
52. An Overview of Relativistic Particle Pushers and their Extension to Arbitrary Order Accuracy, otwierano: czerwca 26, 2026, <https://arxiv.org/html/2603.06509v1>
53. Particle Push in Magnetic Field (Boris Method), otwierano: czerwca 26, 2026, <https://www.particleincell.com/2011/vxb-rotation/>
54. differential-equations: High-Performance ODE/DDE/SDE Solvers in Rust - Reddit, otwierano: czerwca 26, 2026, https://www.reddit.com/r/rust/comments/1kl8b0i/differentialequations_highperformance_odeddesde/
55. Announcing `numr`: A "Batteries-Included" Numerical Library for Rust (NumPy + GPU + Autograd), otwierano: czerwca 26, 2026,

<https://users.rust-lang.org/t/announcing-numr-a-batteries-included-numerical-library-for-rust-numpy-gpu-autograd/138053>

56. differential_equations - Rust - Docs.rs, otwierano: czerwca 26, 2026,
<https://docs.rs/differential-equations>